

PROJECT VICTORY: CHECKOUT EXPERIENCE OPTIMIZATION

Data-driven rollout strategy for Variant B.

BUSINESS GOALS & KPI FRAMEWORK

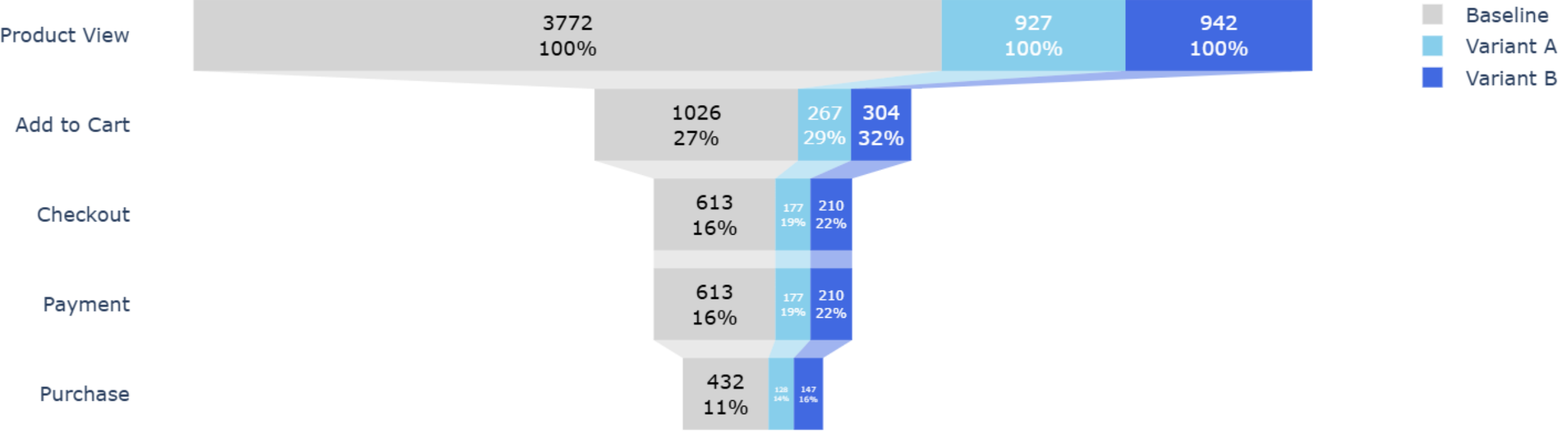
- **The Mission:** To modernize the checkout experience and reduce friction in the final purchase steps.

KPI Category	KPI Name	Definition	Business Purpose
North Star	Checkout Completion Rate (CCR)	Total Purchases / Sessions with Begin Checkout	This metric isolates the checkout experience.
Primary	Session Conversion Rate (CVR)	Total Purchases / Total Eligible Sessions	High-level success metric for the entire session.
Financial	Revenue per Eligible Session (RPES)	Net Revenue / Total Eligible Sessions	The "Bottom Line" used for 30-day impact forecasting.
Basket	Average Order Value (AOV)	Total Net Revenue / Total Orders	Ensures Variant B doesn't lower total spend per user.
Efficiency	Funnel Step Conversions	{Step } n+1} / {Step } n}	Identifies exact page leaks (Shipping, Payment, etc.).
Technical	Payment Failure Rate	Failed Payment Events / Total Payment Attempts	Guardrail metric to detect technical bugs in Variant B.
Profitability	Gross Margin Proxy	(Net Revenue - Total Product Cost – Shipping cost) / Net Revenue	Ensures growth isn't coming at the cost of profitability.
Incentive	Discount Rate	Total Discount Amount / Gross Revenue	Monitors if Variant B relies too heavily on price cuts.
Experience	Median Time-to-Purchase	Median(Time purchase) – Median(Time checkout)	Measures if the new UI reduces or increases user friction.
Growth	Repeat Purchase Rate	Users with > 1 order / Total users	Long-term check on whether the new UI builds loyalty.

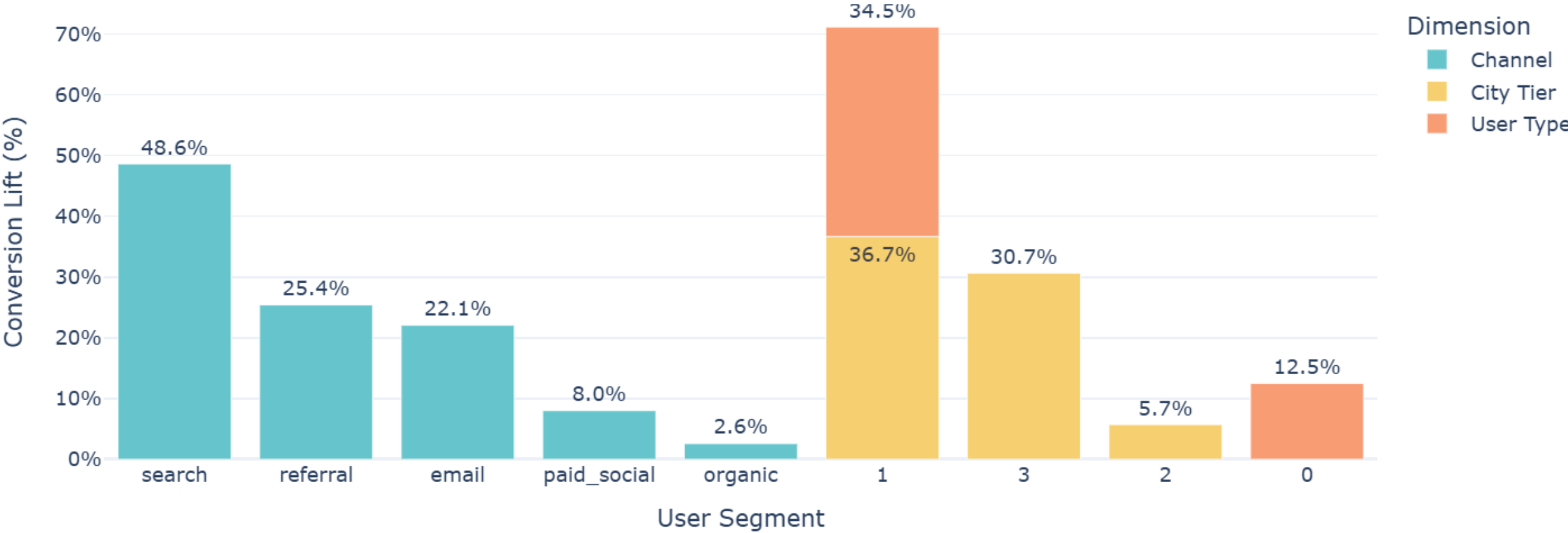
DATA STRATEGY & METHODOLOGY

- Cleaning & Quality: * Applied IQR Capping to handle revenue "whales" without losing data.
- Synchronized event timings to calculate velocity (Time-to-Step).
- Eligibility: Restricted to Web platform and Logged-in users to ensure a clean experiment environment.
- Technical Audit: SRM check passed ($p = 0.37$), confirming a perfectly balanced 50/50 traffic split.

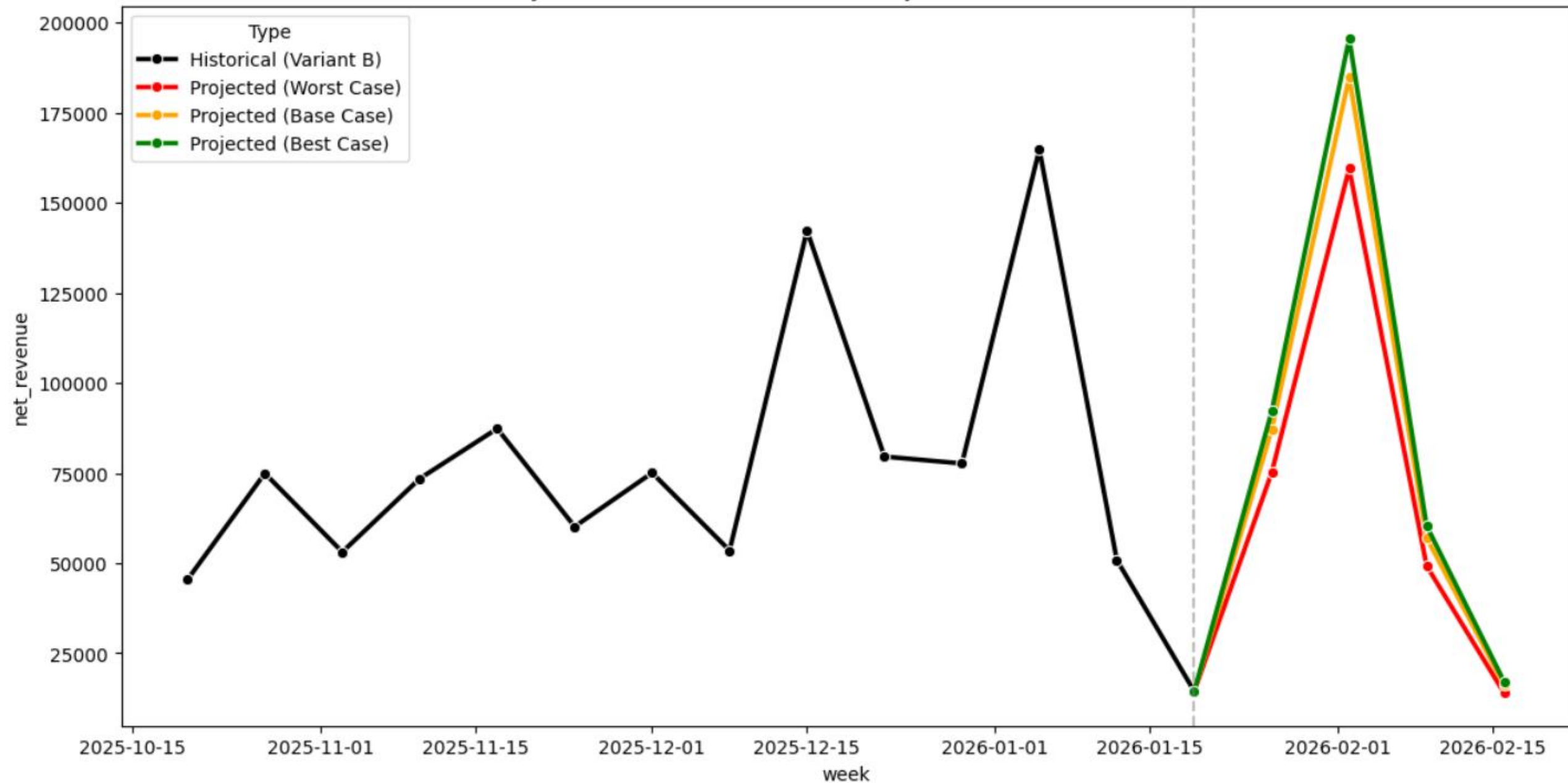
Comparative Analysis: Baseline vs Variants A vs B



HTE Analysis: Where does Variant B Win? (Lift % by Segment)



30-Day Seasonal Revenue Forecast (Projected from Variant B Trends)



KEY INSIGHTS & EXPERIMENT OUTCOME

- The Verdict: 🟠 Directional Winner (Iterate).
- Key Findings:
 - The Funnel Win: Variant B reduced "Product View to Cart" drop-off by 3.5%.
 - The Trust Factor: Users spend more time in Variant B (376s vs 347s), but convert at a higher rate—indicating higher intent.
 - The Latency Risk: A statistically significant speed penalty of 28.5s ($p = 0.035$) is the only barrier to full significance.
 - Stability: 100% Payment Success Rate across all methods confirms technical robustness.
 - Weekly Resilience: Variant B was less volatile during February traffic dips compared to the Baseline.

SEGMENT DEEP DIVE (HTE)

- The Winners:
 - Search Traffic (+48.6% Lift): Variant B is a massive "closer" for users who know exactly what they want.
 - New Users (+34.5% Lift): The new UI creates immediate trust with first-time visitors.
- The Friction Point:
 - Returning Users: Showed neutral/negative sentiment. Likely due to "Change Aversion" where users are frustrated that their "old way" of checking out has changed.

30-DAY FINANCIAL IMPACT ESTIMATE

- The Forecast:Applying current trends to the next 30 days of traffic.
- Assumptions: Includes 5% sensitivity margin and seasonal trend alignment from Variant B.

Scenario	Revenue Impact	Incremental Orders	Confidence
Best Case	+₹249,012	+34	High (Search Segments)
Base Case	+₹127,768	+17	Medium (Average)
Worst Case	-₹86,302	-12	Low (Total Latency fail)

RECOMMENDATIONS & FOLLOW-UPS

- **MoSCoW Prioritization:**

- **MUST (P0): Technical Latency Sprint.** Engineering must reduce the 28.5s UI delay. This is the "blocker" to ₹249k.
- **SHOULD (P1): Segmented Rollout.** Enable Variant B for 100% of **Search** and **New User** traffic immediately.
- **COULD (P2): Sticky "Add to Cart" Button.** Experiment with a sticky CTA to bridge the 67% top-of-funnel leak.
- **COULD (P2): Trust Badges.** Add SSL/Security badges near the final CTA to reduce the 3.2% final-step drop-off.
- **WON'T (For Now):** 100% Rollout for returning users until the speed penalty is resolved.

- **Risks & Limitations:**

- **Limitations:** Small sample size in Tier 3 cities makes that specific lift less certain.
- **Follow-up:** 14-day validation test post-speed optimization to confirm the CVR lift becomes statistically significant ($p < 0.05$).